Summer Heat: Making a Consistent Health Impact

Recently, one study demonstrated the rise of 20% all-cause mortality at temperatures ≥40°C. The study also concluded the spatial variation of all-cause mortality at zonal level with temperature.[1] Global public health scenario is changing because of remarkable burden of diseases and mortality due to climate change, especially extreme temperatures. Studies have shown that heat waves are the most significant weather-related cause of mortality throughout the world. Evidence-based information from all over the globe revealed that climate change has augmented the threat of temperature-related morbidity and mortality, especially in urban areas.[2] Indian cities also experienced heat waves with a significant mortality rate.[1]

It is widely accepted that climate change is occurring and that it is caused mainly by increased emissions of anthropogenic greenhouse gases, particularly over the last few decades. Not only has the average global surface temperature increased, but the frequency and intensity of temperature extremes have also changed.[3]

High temperature has been associated with significant health impacts. The effect of ambient temperature on morbidity is a significant public health issue. Every year, a large number of hospitalizations are associated with exposure to extreme ambient temperatures, especially during summer period in north and central India. To estimate the impact of high temperatures and heat waves on morbidity and mortality in north central India, we explored and documented the morbidity and mortality data at a tertiary hospital during April to July 2016. We used Poisson regression models to estimate the association between daily maximum temperature and rates of all-cause hospital visits and admissions and all-cause mortality. Total admissions were considered and were not identified as either a primary or secondary discharge diagnosis. The associations between heat and specific causes of death were not considered as there were insufficient number of deaths. Accordingly, the goals of this study were to (a) quantify the association between maximum temperature and hospital visits and mortality in a tertiary hospital of a city in North Central India and (b) identify characteristics that may place individuals at greater risk of adverse health effects, We evaluated these goals using population-based data from the hospital covering a population of 50 lakh of 12 districts in three states that have a climate consisting of four seasons with an average temperature of 39.23 ± 3.8°C during summer seasons lasting approximately 135 days.[4]

Mean maximum temperature during summer was 39.23 ± 3.8°C (range: 45.3–33.4°C) in 2016 comprising 135 summer days. Total outpatient department (OPD) patients was 522683 in 2016, and the mean number of OPD patients per day at a daily maximum temperature of <38°C was 1365 which increased to 1546 during the daily maximum temperature of ≥39°C. Mean daily all-cause OPD patients showed a significant association with an increase of 12.5% ($P < 0.001$).

Inpatient department patients (IPD) per day increased at a rate of 16.6% from 208 to 251 when temperature increased to ≥40°C ($P = 0.001$), and deaths per day at a daily maximum temperature of <38°C was 9.8 which increased to 12.5 during the daily maximum temperature of ≥40°C. This shows that 33.3% deaths per day increased when the temperature reached ≥40°C. Mean daily all-cause mortality showed a significant association with daily maximum temperature ($P < 0.001$) [Table 1]. We found increasing morbidity and mortality with relation to rising temperature and conclude that temperature is an important factor affecting all-cause hospital visits in Gwalior city. The association with all-cause hospital admissions was the strongest among males of all age groups of socially deprived and poor, especially belonging to urban areas. This also augments the previous findings and is consistent with results obtained in other areas. All-cause mortality is also correlated with the heat index consistent with various studies.[5,6] In addition, urban residents may be exposed to higher temperatures than residents of surrounding suburban and rural areas because of the "heat island effect" resulting from high thermal absorption by dark paved surfaces and buildings, heat emitted from vehicles and air conditioners, lack of vegetation and trees, and poor. Because of the urban heat island effect, people in urban areas are usually at an increased risk of morbidity from ambient heat exposure.[3]

Conclusions

Prolonged exposure to extreme heat can cause heat exhaustion, heat cramps, heat stroke, and death, as well as exacerbate preexisting chronic conditions, such as various respiratory, cerebral, and cardiovascular diseases. These serious health consequences usually affect more vulnerable populations such

Table 1: Comparison of OPD, IPD, and mortality with temperature variation

Temperature variations Mean±SD [Total days]	OPD/Day (*N*=522683) *n* (%; 95% CI)	IPD/Day (*N*=82043)	Mortality/Day (*N*=3941)
≥39.23±3.8°C[135]	1546.1	251.8	12.5
<38°C±5.2°C[230]	1365.0	208.9	9.8
% difference	12.5%	18.6%	33.3%
P-value	0.0001*	0.001*	0.001*

OPD: Outpatient department, IPD: Indoor patients department (Hospital admissions), *Significant statistically

as the elderly, children, and those with existing cardiovascular and respiratory diseases. Socioeconomic factors, such as economically disadvantaged and socially isolated individuals, are at an increased risk from heat-related burdens. As global temperatures rise and extreme heat events increase in frequency due to climate change we can expect to see more heat-related illnesses and mortality. All other factors held constant, our findings suggest that the current population of northcentral India would experience substantially higher morbidity and mortality if maximum daily temperatures increase from 38°C. National, state, and local public health and emergency management agencies need a clear understanding of the health risks posed by excess heat both now and under the warmer temperatures projected for the future.[7]

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Conflicts of interest

There are no conflicts of interest.

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